



COE-49413 – Computer Vision Semester Project

The semester project is designed to assess your understanding of the course material and your ability to apply computer vision techniques to real-world problems. Working in **groups of 3** is expected. Select a project that is sufficiently scoped for completion within a 4-week timeframe. You may select one of the following options:

1. **Reproduce and Adapt:** Select a state-of-the-art research paper from a recent computer vision conference (see Appendix A). Publicly available project code for most of the papers can be accessed from [here](#). Apply the method to a new application domain (i.e. different dataset) and evaluate its performance.
2. **Extend and Innovate:** Choose a state-of-the-art research paper from a recent computer vision conference (see Appendix A). Utilize the publicly available project code and propose an extension to the method. Evaluate its performance against the original approach.
3. **Solve a Local Problem:** Identify a local computer vision problem relevant to AUS or UAE and develop a solution and evaluate its performance.
4. **Ensemble Learning:** Combine multiple computer vision methods to enhance the accuracy of a specific task. Implement the ensemble approach and evaluate its performance against individual models.

Your project will include some or all the following steps:

- Data Indigestion: data cleaning, and preprocessing to make the data suitable for training.
- Data augmentation: artificially increasing the size of the training dataset by applying various transformations to the existing images.
- Deep learning model: deciding whether to leverage pre-existing deep learning models or create a new one.
- Performance metrics for evaluation: defining appropriate performance metrics for evaluating the performance.

Project Deliverables:

Final project report around 6 pages in length highlighting the methodology used, results and discussion. Use latex template for the report <https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhnncsfqn>

Submit code to GitHub. Make sure the GitHub Readme explains how to run the code step by step for both training and testing cases

Criteria	Description	Weightage
Problem Identification and	• problem complexity and relevance • clear problem statement and justification	15%

literature review	• Comprehensive literature review	
Methodology	• justification of Computer vision techniques applied • quality of implementation and code of GitHub • quality of the readme file • Innovative approach	30%
Result and analysis	• appropriate performance metrics • Comprehensive comparative evaluation with existing methods or between approaches	25%
Organization and structure of report	• Clear delivery of key findings, methodology, results and discussion	20%
Demo	• Quality of the demo	10%

Due: Week 14 (27 April)

Appendix A:

List of Computer Vision Conferences:

CVPR
ICCV
ECCV
ICML
NIPS
AAAI
ACCV
BMVC
ICPR

<https://openaccess.thecvf.com/menu>

<https://paperswithcode.com/area/computer-vision>

<https://arxiv-sanity-lite.com/>